Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item

1. Which function is used to perform a backward pass in PyTorch? 1 / 1 point

- (x) backward()
- ( ) forward()
- ( ) autograd()
- ( ) tensor()

Correct
Correct! The backward() function in PyTorch is used to perform a backward pass to compute the gradient of a tensor.

2. Which of the following steps are involved in building a simple computational graph in PyTorch? 1 / 1 point

- [ ] Initialize weights as zeros
- [x] Perform operations on tensors

Correct
Correct! Performing operations on tensors forms the nodes in the computational graph.

- [ ] Save the model in .pt format
- [x] Use the backward() function to compute gradients

Correct
Correct! Using the backward() function computes the gradients, which is a crucial step in training neural networks.

- [x] Define tensors with requires_grad=True

Correct
Correct! Setting requires_grad=True allows PyTorch to track operations on tensors, which is essential for building the computational graph.

3. Which of the following hyperparameters can impact the performance of a neural network model? 1 / 1 point

- [x] Learning rate

Correct
Correct! The learning rate is crucial as it determines the step size during gradient descent.

- [x] Number of layers

Correct
Correct! The number of layers can affect the model's capacity to learn and generalize.

- [x] Batch size

Correct
Correct! The batch size can affect the stability and efficiency of training.

- [ ] Output data type
- [x] Activation function

Correct
Correct! The choice of activation function can affect the non-linearity and learning capability of the model.

4. Which of the following steps are involved in the training loop of a PyTorch model? 1 / 1 point

- [x] Optimizer step

Correct
Correct! The optimizer step updates the model parameters based on the calculated gradients.

- [ ] Parameter initialization
- [x] Forward pass

Correct
Correct! The forward pass involves passing the input data through the model to obtain predictions.

- [x] Backward pass

Correct
Correct! The backward pass involves calculating the gradients of the loss with respect to the model parameters.

- [ ] Model evaluation

5. Which component in a PyTorch model is responsible for updating the model parameters based on the calculated gradients? 1 / 1 point

- (x) Optimizer
- ( ) Loss function
- ( ) DataLoader
- ( ) Activation function

Correct
Correct! The optimizer updates the model parameters based on the calculated gradients and the learning rate.